Unit 1 - Math Fundamentals
1.2 Sets

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Outline

1. Why Sets?
2. Sets and Membership
3. Subsets
4. Set Operations
5. Venn Diagrams and Counting
6. Sets of Numbers
7. Intervals
8. Ordered Pairs and Tuples
9. Review

Why Sets?

A Problem to Motivate Us

A question about a patient population

"How many patients have diabetes **or** *hypertension, but* **not both***?"*

To answer this we need language for:

- a **collection** of patients ("the diabetics"),
- **combining** collections ("diabetic or hypertensive"),
- **removing** one collection from another ("but not both"),
- and **counting** what is left.

That language is **set theory**. It is the same logic as Unit 1-1, AND, OR, NOT, but now applied to collections of things rather than to statements.

Where This Shows Up in Data Science

Sets are everywhere once you look

- **Cohorts and subgroups:** every "group of patients" is a set.
- **Joining data:** merging two tables is an intersection of their keys.
- **Domains of functions:** the set of inputs a function accepts.
- **Reference ranges:** "normal" lab values are an **interval**, a set of numbers.
- **Probability:** an event is a set of outcomes (Unit 4).

What we will build

1. **Sets**: membership, and **subsets**.
2. The **operations**: union, intersection, complement.
3. **Venn diagrams** and counting.
4. The standard **sets of numbers** and **intervals**.
5. **Ordered pairs** and tuples, where sets meet coordinates.

Sets and Membership

What Is a Set?

Set

A **set** is a collection of objects, called its **elements** (or **members**).

- Sets are usually named with capital letters: A, B, C.
- Elements are listed inside curly braces:

$$A = \{1, 2, 3\}$$

- **Order does not matter**: $\{1, 2, 3\}$ and $\{3, 1, 2\}$ are the same set.
- **Repeats do not matter**: $\{1, 1, 2\}$ is just $\{1, 2\}$.

A set is defined entirely by *what is in it*, nothing else.

Membership

To say that an object is in a set, we use the symbol $\in$:

$$1 \in A \qquad \text{“1 is an element of } A\text{”}$$

And to say it is not, we strike it through:

$$5 \notin A \qquad \text{“5 is not an element of } A\text{”}$$

Connection to Unit 1-1

“$x \in A$” is a **statement**, it is true or false, and so it can be combined with $\neg$, $\wedge$, $\vee$ like any other. Notice that its truth depends on which x you pick, which is exactly what set-builder notation will use. Everything we do with sets will mirror something we did with logic.

Some Special Sets

Singletons

A set with exactly one element, e.g. $\{1\}$, is called a **singleton**.

Note that 1 and $\{1\}$ are different things: a number, and a box containing that number.

The empty set

The set with **no** elements is called the **empty set**:

$$\emptyset = \{\ \}$$

A universe

Often we fix a **universal set** U containing everything under discussion, e.g. all patients in the study. "Everything else" always means "everything else in U."

Set-Builder Notation

Listing elements only works for small sets. For large or infinite ones, we describe the **rule** that members satisfy:

$$A = \{\, x \mid x > 0 \,\} \quad \text{“the set of all positive numbers”}$$

Read the vertical bar (pipe) as **“such that”**:

“the set of all x *such that* x is greater than 0”

More examples

- $\{\, x \mid x$ is a day of the week $\}$
- $\{\, x \mid x$ is a patient and x is diabetic $\}$
- $\{\, x \in U \mid x$ is diabetic $\}$ (membership can be stated up front)

The rule after the bar is a **statement about** x, logic again.

Exercise: Translating Set-Builder Notation

Part 1, translate set-builder notation into plain English

1. $\{\, x \mid x < 0 \,\}$
2. $\{\, x \in U \mid x$ is pregnant $\,\}$ ($U =$ all patients in the study)
3. $\{\, x \mid x$ is a month with exactly 30 days $\,\}$, then list it.

Part 2, now go the other way, English into set-builder notation

1. The set of all patients who are diabetic **and** over 65.
2. The set of all letters that are vowels.
3. The set of all numbers between 10 and 20.

Exercise: Solution

Part 1, into English

1. "the set of all x *such that* x is less than 0", the negative numbers.
2. "the set of all patients in the study who are pregnant."
3. "the set of all months with exactly 30 days"
   $= \{\text{April}, \text{June}, \text{September}, \text{November}\}$.

Part 2, into set-builder

1. $\{\, x \in U \mid x \text{ is diabetic } \wedge\ x \text{ is over } 65 \,\}$ (the AND from Unit 1-1)
2. $\{\, x \mid x \text{ is a vowel} \,\} = \{a, e, i, o, u\}$
3. $\{\, x \mid 10 < x < 20 \,\}$

That last one is slippery: "between" never says whether 10 and 20 themselves count. **Intervals** will pin this down.

Cardinality

Cardinality

The **cardinality** of a set A, written $|A|$, is the number of elements it contains.

Examples

- $A = \{1, 2, 3\} \Rightarrow |A| = 3$
- $|\{7\}| = 1$
- $|\emptyset| = 0$

Every "how many patients . . ." question is a cardinality question. We will come back to this when we count with Venn diagrams.

Exercise

Try it

Let

$$A = \{2, 4, 6, 8\}$$
$$B = \{\, x \mid x \text{ is a letter in the word “level”} \,\}$$

1. Write B by listing its elements.
2. Find $|A|$ and $|B|$.
3. True or false: $4 \in A$, $\{4\} \in A$, $5 \notin A$.

Exercise: Solution

1. Listing B

The letters of "level" are l, e, v, e, l, but **repeats do not matter**, so $B = \{\mathsf{l}, \mathsf{e}, \mathsf{v}\}$.

2. Cardinality

$|A| = 4$ and $|B| = 3$, three *distinct* letters, not five.

3. True or false

- $4 \in A$, **true**, 4 is one of the listed elements.
- $\{4\} \in A$, **false**. The elements of A are numbers; $\{4\}$ is a *set*, and no set appears in A.
- $5 \notin A$, **true**, 5 is not listed.

Subsets

Subsets

Subset

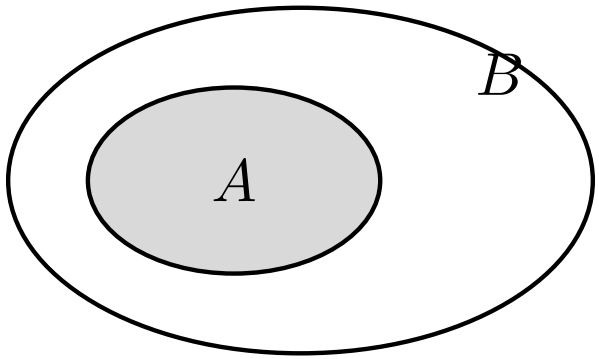
A is a **subset** of B, written $A \subseteq B$, if **every** element of A is also an element of B.

Written with the implication from Unit 1-1, for every x:

$$x \in A \quad \rightarrow \quad x \in B$$

Example

$\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2, 3\} \subseteq \{1, 2, 3\}$, a set is a subset of itself.



Proper Subsets, and the Empty Set

Proper subset

A is a **proper** subset of B, written $A \subset B$, if $A \subseteq B$ **and** $A \neq B$. That is, B has at least one element that A does not.

$A \subseteq B$	contained in, possibly equal
$A \subset B$	contained in, definitely smaller

Convention: $\emptyset \subseteq B$ for every set B

The empty set is a subset of **everything**.

Why? To fail, $\emptyset$ would need an element that is not in B, and it has no elements at all to offer. The condition is **vacuously true** (Unit 1-1).

Set Equality

Two sets are equal when each contains the other

$$A = B \quad \text{means} \quad A \subseteq B \ \text{ and } \ B \subseteq A$$

This is the set-theory version of the bi-implication $\leftrightarrow$ from Unit 1-1: $x \in A$ if and only if $x \in B$.

This is also a proof strategy

To prove two sets are the same, prove **both** containments:

1. Take an arbitrary $x \in A$ and show $x \in B$.
2. Take an arbitrary $x \in B$ and show $x \in A$.

Exercise

Try it

Let $A = \{1, 2, 3\}$.

1. List **all** subsets of A. How many are there?
2. Which of those are **proper** subsets?
3. True or false: $\emptyset \subseteq A$, $\emptyset \in A$, $A \subseteq A$, $A \subset A$.

Exercise: Solution

1. All subsets, eight of them

$$\emptyset, \quad \{1\}, \{2\}, \{3\}, \quad \{1,2\}, \{1,3\}, \{2,3\}, \quad \{1,2,3\}$$

A set with n elements has 2^n subsets, here $2^3 = 8$.

2. Proper subsets

Every one of them except $\{1,2,3\}$ itself, **seven**.

3. True or false

- $\emptyset \subseteq A$, **true**. The empty set is a subset of everything.
- $\emptyset \in A$, **false**. The elements of A are $1, 2, 3$; $\emptyset$ is not one of them.
- $A \subseteq A$, **true**. $A \subset A$, **false**: proper demands $A \neq A$.

The Power Set

The eight subsets we just listed are objects in their own right, so we can collect them into a set.

Power set

The **power set** of A, written $\mathcal{P}(A)$, is the set of **all** subsets of A:

$$\mathcal{P}(A) = \{\, S \mid S \subseteq A \,\}$$

Its elements are themselves **sets**, not the original elements.

Example: $A = \{1, 2, 3\}$

$$\mathcal{P}(A) = \big\{\, \emptyset,\ \{1\}, \{2\}, \{3\},\ \{1,2\}, \{1,3\}, \{2,3\},\ \{1,2,3\} \,\big\}$$

$|\mathcal{P}(A)| = 8$, and both $\emptyset$ and A itself are members.

Why 2^n? One yes/no choice per element, in or out, gives 2^n subsets, the same count as the rows of an n-variable truth table (Unit 1-1). **Watch the symbols:** $\{1\} \subseteq A$, but $\{1\} \in \mathcal{P}(A)$.

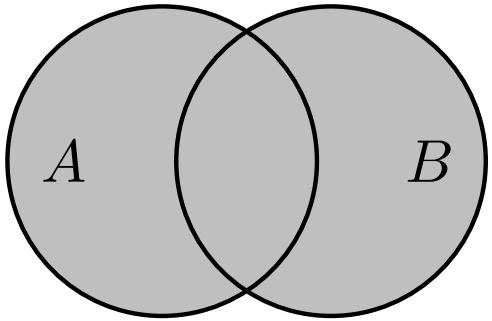
Set Operations

Union

Union

The **union** of A and B, written $A \cup B$, is the set of everything in A, in B, or in both:

$$A \cup B = \{\, x \mid x \in A \ \ \textbf{or} \ \ x \in B \,\}$$



$\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}$

Note

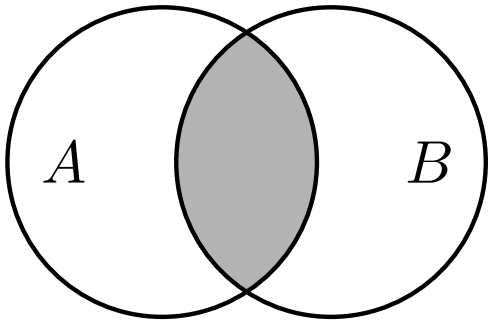
Union is the set version of the logical **OR**, and it is the *inclusive* OR: elements in both sets are included, listed once.

Intersection

Intersection

The **intersection** of A and B, written $A \cap B$, is the set of everything in **both**:

$$A \cap B = \{\, x \mid x \in A \ \textbf{ and } \ x \in B \,\}$$



$\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}$

Note

Intersection is the set version of the logical **AND**.

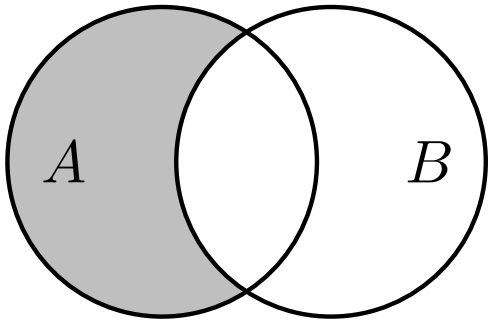
If $A \cap B = \emptyset$, the sets share nothing and are called **disjoint**.

Difference (Relative Complement)

Set difference

The **difference** $A \setminus B$ is everything in A that is **not** in B:

$$A \setminus B = \{\, x \in A \mid x \notin B \,\}$$



Note

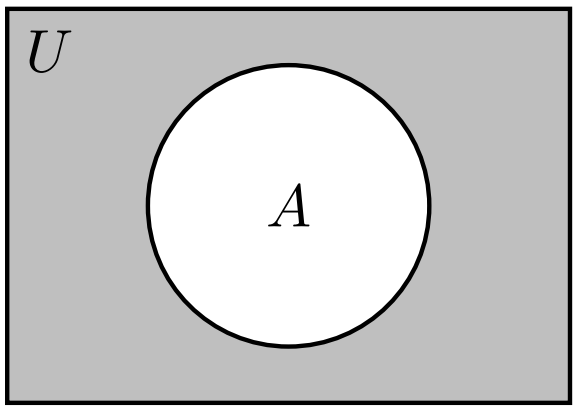
Difference is the set version of **AND NOT**. **Order matters**:

$$\{1, 2, 3\} \setminus \{2, 3, 4\} = \{1\}$$
$$\{2, 3, 4\} \setminus \{1, 2, 3\} = \{4\}$$

Complement

When a universal set U is fixed, the **complement** of A is everything in U outside of A:

$$\bar{A} = U \setminus A = \{ x \in U \mid x \notin A \}$$



Note

Complement is the set version of **NOT**.

It only makes sense once U is agreed upon, "not diabetic" means nothing until we say "among whom."

Some texts write A' instead of $\bar{A}$.

Worked Examples

Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$.

- $A \cup B = \{1, 2, 3, 4\}$
- $A \cap B = \{2, 3\}$
- $A \setminus B = \{1\}$
- $B \setminus A = \{4\}$

Exercise

Try it

With $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{3, 4, 5\}$, find:

$$A \cap C, \qquad (A \cup B) \cap C, \qquad A \setminus (B \cap C)$$

Exercise: Solution

$A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$. Work from the inside of the parentheses outward.

$A \cap C$

Elements in both A and C. Only 3 qualifies: $A \cap C = \{3\}$

$(A \cup B) \cap C$

First the inside: $A \cup B = \{1, 2, 3, 4\}$.

Then intersect with $C = \{3, 4, 5\}$: $(A \cup B) \cap C = \{3, 4\}$

$A \setminus (B \cap C)$

First the inside: $B \cap C = \{3, 4\}$.

Then remove those from $A = \{1, 2, 3\}$, the 4 was not there to remove: $A \setminus (B \cap C) = \{1, 2\}$

Properties of Union and Intersection

These behave much like $+$ and $\times$ in ordinary algebra.

Commutative, order does not matter

$$A \cup B = B \cup A \qquad A \cap B = B \cap A$$

Associative, grouping does not matter

$$A \cup (B \cup C) = (A \cup B) \cup C \qquad A \cap (B \cap C) = (A \cap B) \cap C$$

$\cup$ and $\cap$ are **binary**, each joins two sets at a time. Associativity is what lets us drop the parentheses and write $A \cup B \cup C$, exactly as for $1 + 2 + 3$.

Distributive, each distributes over the other

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

The first of these has no arithmetic analogue. $+$ does not distribute over $\times$.

De Morgan's Laws for Sets

The laws from Unit 1-1 carry over word for word, with $\cup$ for OR and $\cap$ for AND.

De Morgan's Laws (sets)

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$
$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Why they are the same laws

$$\begin{aligned} x \in \overline{A \cap B} &\Leftrightarrow \neg\big(x \in A \ \wedge \ x \in B\big) \\ &\Leftrightarrow \neg(x \in A) \ \vee \ \neg(x \in B) \qquad \text{(logical De Morgan)} \\ &\Leftrightarrow x \in \bar{A} \cup \bar{B} \end{aligned}$$

Set theory is logic with the statements collected into piles.

Back to the Cohort

Unit 1-1 opened with a criterion we could write in logic but not yet *collect*:

The English

"Patients who are diabetic AND (hypertensive OR obese), but NOT pregnant."

Let U be all patients in the study, and name each property as a statement about x:

$$d(x) = \text{“}x \text{ is diabetic”} \qquad h(x) = \text{“}x \text{ is hypertensive”}$$
$$o(x) = \text{“}x \text{ is obese”} \qquad g(x) = \text{“}x \text{ is pregnant”}$$

The cohort is a set

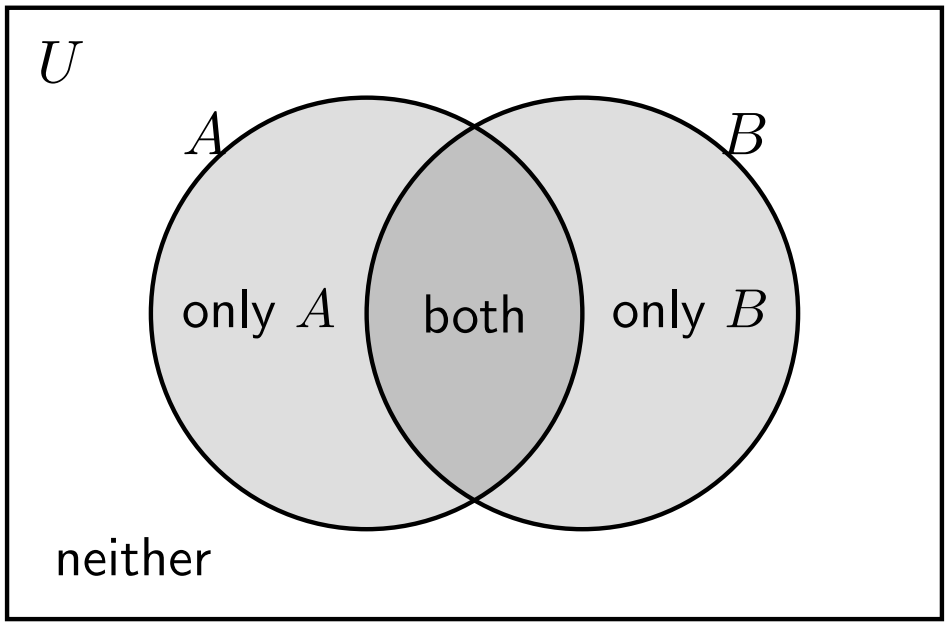
$$\text{Cohort} \;=\; \left\{ x \in U \;\middle|\; d(x) \wedge (h(x) \vee o(x)) \wedge \neg g(x) \right\}$$

Logic supplies the rule; set-builder notation turns it into a **collection you can count**.

Venn Diagrams and Counting

Reading a Two-Set Venn Diagram

Two overlapping circles cut the universe into **four** regions, and every patient sits in exactly one of them.



only A	$A \setminus B$
both	$A \cap B$
only B	$B \setminus A$
neither	$\overline{A \cup B} = \bar{A} \cap \bar{B}$

Inclusion-Exclusion

How many elements are in $A \cup B$? Adding $|A|$ and $|B|$ counts the overlap **twice**, so we subtract it back off once:

Inclusion-exclusion (two sets)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example

In a clinic of 100 patients, let D be the set of diabetics and H the set of hypertensives. Suppose $|D| = 40$, $|H| = 55$, and $|D \cap H| = 25$.

$$|D \cup H| = 40 + 55 - 25 = 70$$

So 70 patients have at least one condition, and $100 - 70 = 30$ have neither.

Notice that $40 + 55 = 95 \neq 70$. Forgetting to subtract the overlap is the classic error.

Answering the Opening Question

The question

"How many patients have diabetes **or** *hypertension, but* **not both***?"*

In symbols, that is the region $(D \cup H) \setminus (D \cap H)$, the two crescents, but not the middle:

$$|(D \cup H) \setminus (D \cap H)| \;=\; |D \cup H| - |D \cap H|$$

With the numbers from the previous slide:

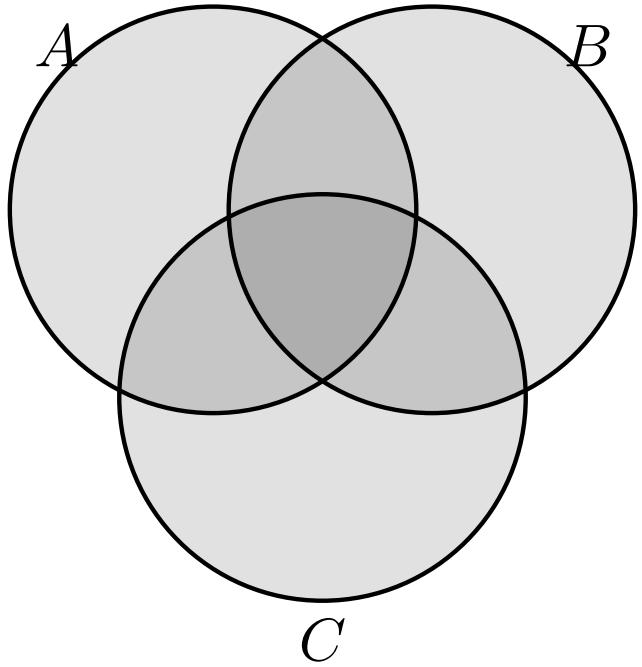
$$70 - 25 = 45$$

Recognize the shape

"Or, but not both" is the **exclusive OR** from Unit 1-1. The same idea, now counted.

Three-Set Venn Diagrams

Three circles cut the universe into **eight** regions (2^3, one for each in/out combination, exactly like a truth table).



Strategy: fill in from the middle outward

Start with $|A \cap B \cap C|$, then the two-way overlaps (subtracting the center), then the single regions. Each region gets counted exactly once.

Exercise

Try it

A clinic surveys 120 patients. Let E be the set who exercise regularly, and D the set who follow a special diet:

$$|E| = 70, \qquad |D| = 45, \qquad |E \cap D| = 25$$

1. How many do **at least one** of the two?
2. How many do **neither**?
3. How many exercise but do **not** diet?

Exercise: Solution

1. At least one, inclusion-exclusion

$$|E \cup D| = |E| + |D| - |E \cap D| = 70 + 45 - 25 = 90$$

2. Neither, everything outside that union

$$|\overline{E \cup D}| = 120 - 90 = 30$$

3. Exercise but not diet, a difference

$$|E \setminus D| = |E| - |E \cap D| = 70 - 25 = 45$$

The 25 who do both are already counted inside E, so subtract them off.

Check the four regions: $45 + 25 + 20 + 30 = 120$, exercise only, both, diet only, neither.

Sets of Numbers

The Standard Number Systems

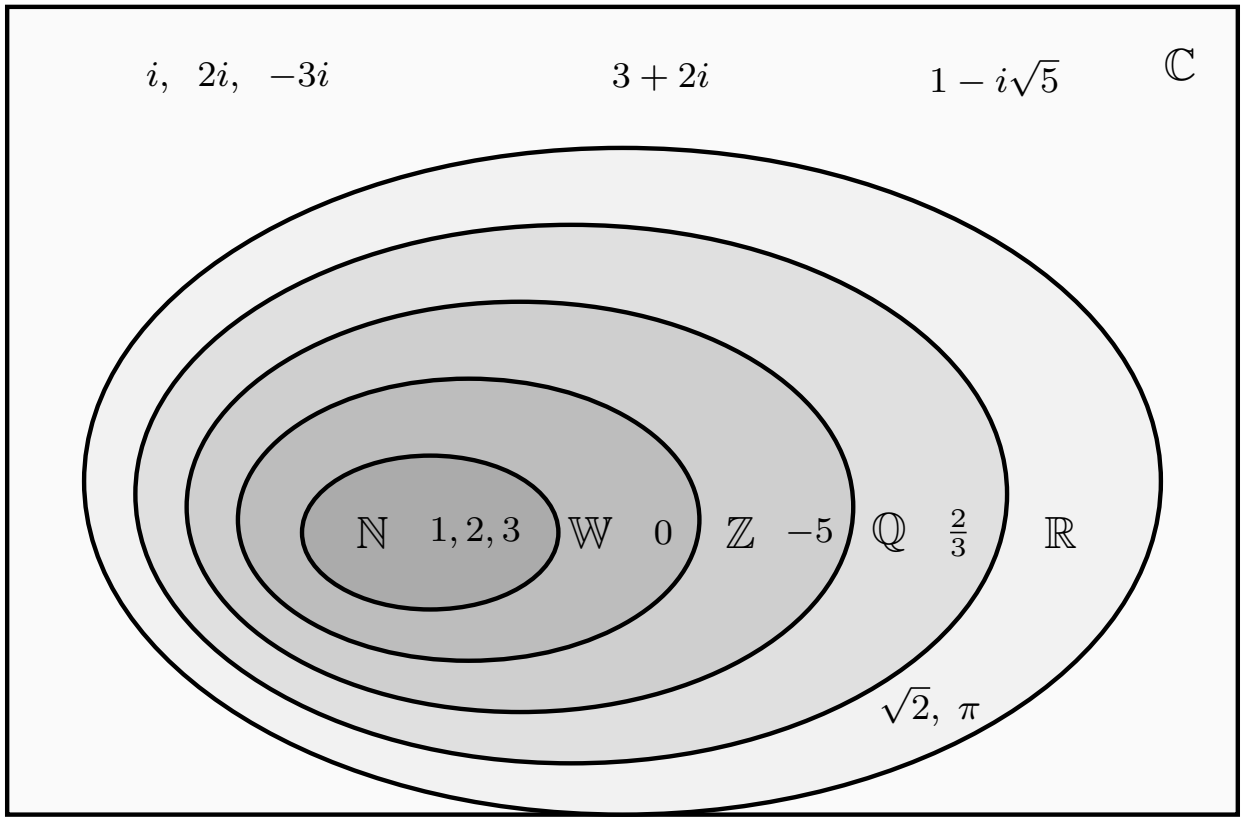
- **Natural numbers:** $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$, the counting numbers.
- **Whole numbers:** $\mathbb{W} = \{0, 1, 2, 3, 4, \ldots\}$, the naturals, plus zero.
- **Integers:** $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$, the wholes, plus negatives.
- **Rational numbers:** $\mathbb{Q} = \left\{ \frac{p}{q} \;\middle|\; p, q \in \mathbb{Z},\ q \neq 0 \right\}$, ratios of integers.
- **Real numbers:** $\mathbb{R}$, every rational **and** every irrational.
- **Complex numbers:** $\mathbb{C} = \{\, a + bi \mid a, b \in \mathbb{R} \,\}$, a real part plus an imaginary part, where $i = \sqrt{-1}$.

Each one sits inside the next

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

The imaginary numbers are **not** a link in that chain: no real number squares to a negative, so $2i \notin \mathbb{R}$. They sit inside $\mathbb{C}$ alongside $\mathbb{R}$, not around it. ($\mathbb{Z}$ is the German *Zahlen*, "numbers"; $\mathbb{Q}$ is for *quotient*.)

The Number Systems, Nested



Inside $\mathbb{R}$ but outside $\mathbb{Q}$ are the **irrationals**, $\sqrt{2}$, π, which are not ratios of integers. Inside $\mathbb{C}$ but outside $\mathbb{R}$ is everything with a nonzero **imaginary** part: i is nowhere on the real line.

Rational vs. Irrational

A rational number's decimal expansion either stops or repeats

- $\frac{1}{2} = 0.5$ (terminates)
- $\frac{1}{3} = 0.3333\ldots$ (repeats)
- $\frac{1}{7} = 0.142857\,142857\ldots$ (repeats, with period 6)

An irrational number's decimal expansion does neither

- $\sqrt{2} = 1.41421356\ldots$
- $\pi = 3.14159265\ldots$
- $e = 2.71828182\ldots$

A warning that Unit 1-4 will pay off

A computer stores only finitely many digits, so it never holds π, or even $\frac{1}{3}$, exactly. That is the subject of **floating-point arithmetic**.

Exercise

Try it

For each number below, name **every** one of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ that it belongs to, put a check in each box that applies.

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$
7					
-3					
0					
$\frac{5}{8}$					
$\sqrt{9}$					
$\sqrt{5}$					

Exercise: Solution

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	
7	✓	✓	✓	✓	✓	
-3			✓	✓	✓	negative
0		✓	✓	✓	✓	$\mathbb{N}$ starts at 1
$\frac{5}{8}$				✓	✓	a ratio of integers
$\sqrt{9}$	✓	✓	✓	✓	✓	$\sqrt{9} = 3$
$\sqrt{5}$					✓	irrational

The two traps

$\sqrt{9}$ is simply 3, a square root is not automatically irrational. And 0 lands in $\mathbb{W}$ but not $\mathbb{N}$, since we take $\mathbb{N} = \{1, 2, 3, \ldots\}$.

Exercise: Implications Between Number Systems

Try it: true or false? If false, give a counterexample.

1. $x \in \mathbb{N} \rightarrow x \in \mathbb{Z}$
2. $x \in \mathbb{R} \rightarrow x \in \mathbb{Z}$
3. $x \in \mathbb{Z} \rightarrow x \in \mathbb{Q}$
4. $x \in \mathbb{Q} \rightarrow x \in \mathbb{R}$
5. $x \notin \mathbb{Q} \rightarrow x \notin \mathbb{R}$
6. $x \notin \mathbb{Z} \rightarrow x \notin \mathbb{N}$

Recall from Unit 1-1 that an implication fails only when the left side is **true** and the right side is **false**, so one counterexample settles it.

Exercise: Solution

Claim	T/F	Why
1. $x \in \mathbb{N} \to x \in \mathbb{Z}$	T	every natural number is an integer
2. $x \in \mathbb{R} \to x \in \mathbb{Z}$	F	$x = 0.5$ is real but not an integer
3. $x \in \mathbb{Z} \to x \in \mathbb{Q}$	T	$n = \frac{n}{1}$, a ratio of integers
4. $x \in \mathbb{Q} \to x \in \mathbb{R}$	T	every rational sits on the real line
5. $x \notin \mathbb{Q} \to x \notin \mathbb{R}$	F	$x = \sqrt{2}$ is irrational **and** real
6. $x \notin \mathbb{Z} \to x \notin \mathbb{N}$	T	the contrapositive of #1

Notice what these really ask. "$x \in A \to x \in B$" is true for **every** x exactly when $A \subseteq B$, so the true rows are precisely the containments in $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, read left to right. Reading one *right to left* (#2) or negating without flipping the order (#5) breaks it.

How Big Is an Infinite Set?

We defined the **cardinality** $|A|$ as the number of elements in A, so $|\{1, 2, 3\}| = 3$. What, then, is the cardinality $|\mathbb{N}|$?

Claim: no natural number is the cardinality of $\mathbb{N}$

Suppose one were, say $|\mathbb{N}| = N$ for some $N \in \mathbb{N}$.

Then $\mathbb{N} = \{1, 2, 3, \ldots\}$ would stop after N elements, so its **largest** element would be N itself.

But $N \in \mathbb{N}$ forces $N + 1 \in \mathbb{N}$ as well, and $N + 1 > N$. So N is **not** the largest element, which contradicts what we just said.

Nothing about N was special: *whatever* number we pick, adding 1 produces a bigger one still in $\mathbb{N}$. So the assumption fails for every candidate, and $|\mathbb{N}|$ is no natural number at all. ■

Writing $|\mathbb{N}| = \infty$ does not rescue us either, we are about to find that **not all infinities are the same size**. We need a notion of "same cardinality" that never requires finishing a count, yet agrees with ordinary counting on finite sets.

Same Size Means Perfect Pairing

A theater has as many seats as ticket-holders when **everyone sits, every seat is filled, and nobody doubles up**, nothing was counted.

Definition by pairing

Two sets have the **same cardinality** if their elements can be matched one-to-one, with nothing left over on either side.

Same cardinality

$A = \{2, 4, 6\}, \quad B = \{a, b, c\}$

A		B
2	$\longleftrightarrow$	a
4	$\longleftrightarrow$	b
6	$\longleftrightarrow$	c

Nothing left over: $|A| = |B| = 3$.

Different cardinality

$C = \{1, 2, 3, 4\}, \quad D = \{x, y, z\}$

C		D
1	$\longleftrightarrow$	x
2	$\longleftrightarrow$	y
3	$\longleftrightarrow$	z

4 left over

D runs out first, as does every other pairing: $|C| \neq |D|$.

The Unsettling Example

The pairing test never asks us to finish counting, so it still works when both sets are **infinite**. Pair each natural number with its double:

$$\mathbb{N} = \{1, 2, 3, 4, \ldots\} \qquad E = \{2, 4, 6, 8, \ldots\}$$

$$\begin{array}{ccc} 1 & \longleftrightarrow & 2 \\ 2 & \longleftrightarrow & 4 \\ 3 & \longleftrightarrow & 6 \\ 4 & \longleftrightarrow & 8 \\ \vdots & & \vdots \\ n & \longleftrightarrow & 2n \end{array}$$

A part can be as big as the whole

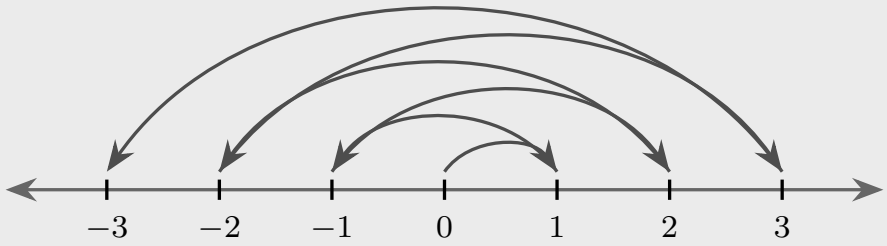
The rule $n \leftrightarrow 2n$ reaches **every** even number, and no natural number is used twice, nothing is left over on either side. So $|\mathbb{N}| = |E|$, even though E is a *proper subset* of $\mathbb{N}$: it is missing every odd number.

For a finite set that is impossible, strip out an element and the count drops. Infinite sets simply do not behave that way.

Countable Infinity: $\aleph_0$

A set is **countable** if its elements can be arranged in an infinite **list** $a_1, a_2, a_3, \ldots$ reaching every one, that is, paired off with $\mathbb{N}$. That size is $\aleph_0$, "aleph-null," the smallest infinity.

The integers: do not start at $-\infty$, zig-zag outward from 0



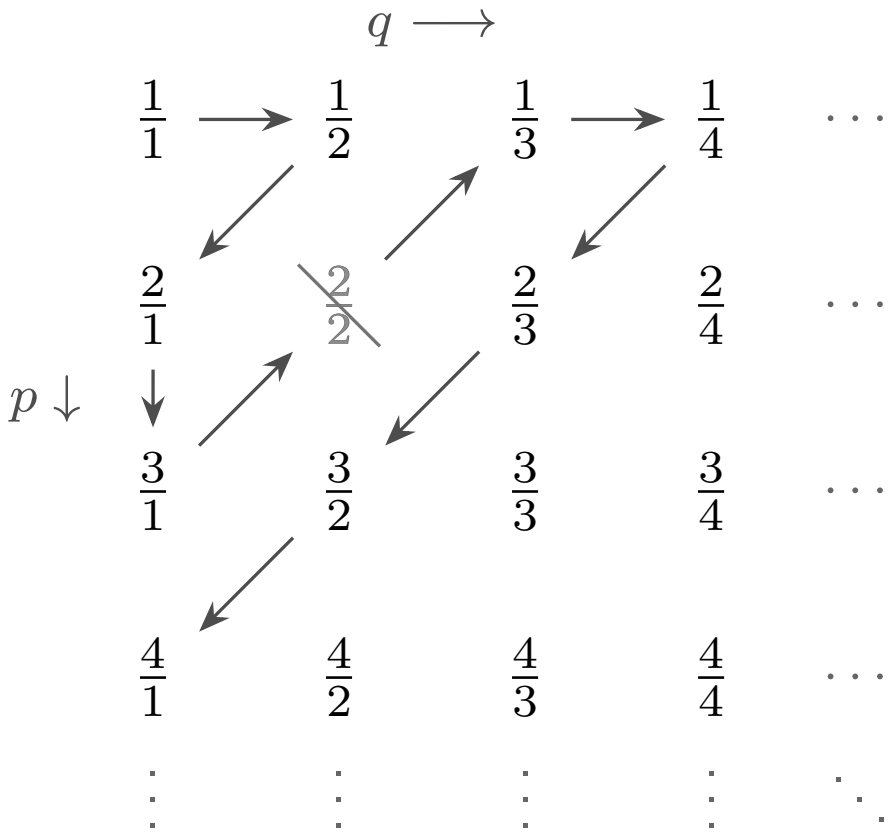
Reading the arcs in order gives the list $0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ \ldots$

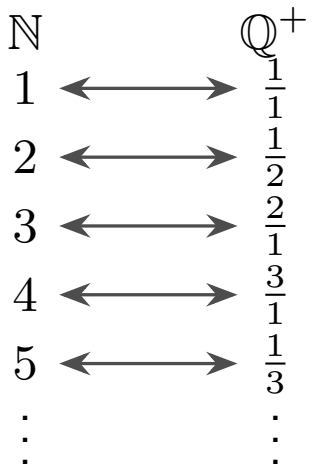
$\mathbb{N}$		$\mathbb{Z}$
1	$\longleftrightarrow$	0
2	$\longleftrightarrow$	1
3	$\longleftrightarrow$	-1
4	$\longleftrightarrow$	2
$\vdots$		$\vdots$

Position n in the list pairs with the nth integer out: every integer is reached, -2 is 5th, -17 is 34th, and none twice. So $|\mathbb{Z}| = \aleph_0$.

The Rationals Are Countable Too

Every positive fraction $\frac{p}{q}$ sits in a **grid**, numerator p picks the row, denominator q the column, and sweeping the **diagonals** threads one path through all of them.





$\frac{2}{2}$ is struck out, it *is* $\frac{1}{1}$. Skipping repeats, the path reaches every positive rational once; interleaving the negatives and 0 as for $\mathbb{Z}$ gives $|\mathbb{Q}| = \aleph_0$.

Cantor's Diagonal Argument

Suppose every real between 0 and 1 could be listed too, in *some* order:

$$
\begin{aligned}
r_1 &= 0.\underline{2}5819340671\ldots \\
r_2 &= 0.4\underline{9}710283561\ldots \\
r_3 &= 0.60\underline{4}82137952\ldots \\
r_4 &= 0.317\underline{6}05942813\ldots \\
r_5 &= 0.8524\underline{0}3718694\ldots \\
r_6 &= 0.19365\underline{7}240817\ldots \\
r_7 &= 0.472018\underline{3}56921\ldots \\
r_8 &= 0.50837619\underline{2}470\ldots \\
r_9 &= 0.2619483\underline{7}5186\ldots \\
r_{10} &= 0.739162840\underline{2}65\ldots \\
&\vdots
\end{aligned}
$$

Build a number the list misses

Walk down the underlined diagonal and replace each digit by **any different** digit, $\underline{2} \to 7$, $\underline{9} \to 3$, $\underline{4} \to 9$, and so on:

$$d \;=\; 0.7391628405\ldots$$

The rows may sit in **any order**: nothing here used one. Shuffle them, re-read the new diagonal, and the recipe still works.

Could d be a line further down, r_{10}, say? Stack them and the first **nine** digits agree:

$$
\begin{aligned}
d &= 0.7391628405\ldots \\
r_{10} &= 0.7391628402\ldots
\end{aligned}
$$

The tenth splits them, since that is the digit the rule changed. Every r_n is defeated the same way at its own nth digit, so d is on **no** line of the list.

What the Diagonal Proves

The contradiction

We assumed the list contained **every** real between 0 and 1. The diagonal hands us one it missed. So no such list exists, and it is not a matter of trying harder, since the recipe defeats *any* list you propose.

The reals are uncountable

$\mathbb{R}$ cannot be paired off with $\mathbb{N}$. Its cardinality is strictly larger, and is written $\mathfrak{c}$, "the continuum":

$$\aleph_0 \;=\; |\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| \;<\; |\mathbb{R}| \;=\; \mathfrak{c}$$

Notice where the jump happens: adding negatives ($\mathbb{Z}$) and all the fractions ($\mathbb{Q}$) costs nothing. Adding the **irrationals** changes the size of infinity.

The Gap Between the Two

An honest open question

Is there a set strictly bigger than $\mathbb{N}$ but strictly smaller than $\mathbb{R}$, some size wedged between $\aleph_0$ and $\mathfrak{c}$?

The guess that there is **no** such size is the **continuum hypothesis**.

The answer is stranger than yes or no

Gödel and Cohen proved that the continuum hypothesis can be neither **proved** nor **disproved** from the standard axioms of set theory. Both answers are consistent. You may assume either one.

A reminder that "how many?" is a deeper question than it looks, and that the number line you have used since grade school still holds open problems.

Intervals

Intervals, and How We Write Them

Interval

An **interval** is the set of **all** real numbers lying between two given numbers a and b, called the **endpoints**. The bracket records whether each endpoint is itself included:

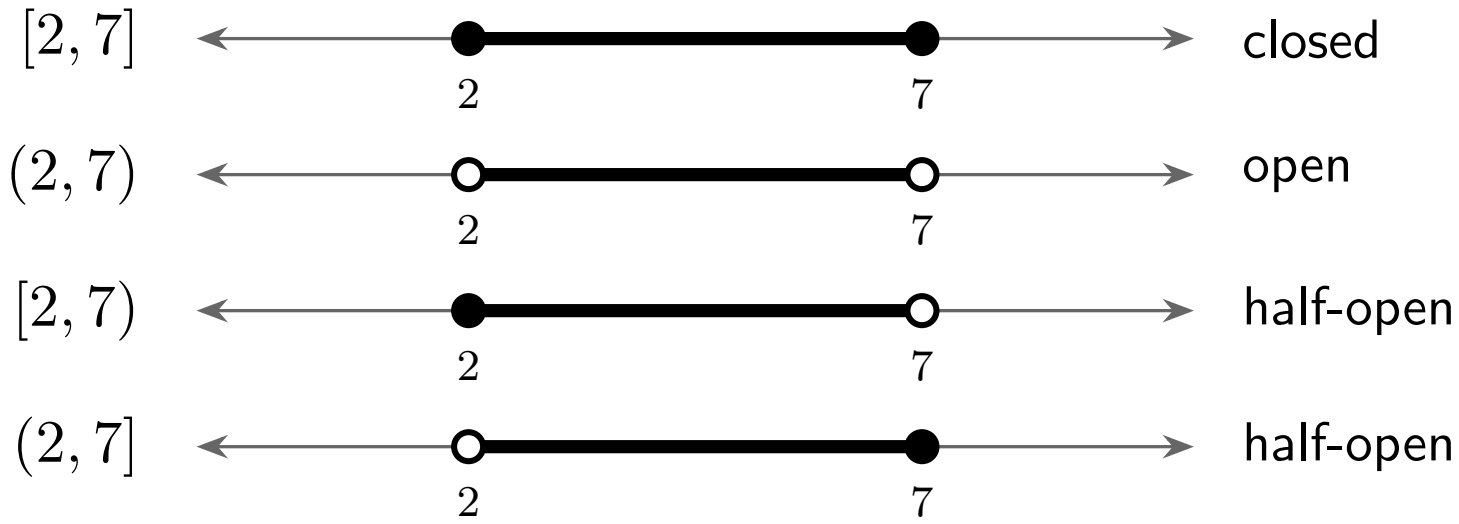
$$\begin{aligned}
[a,b] &= \{\, x \in \mathbb{R} \mid a \le x \le b \,\} \\
(a,b) &= \{\, x \in \mathbb{R} \mid a < x < b \,\} \\
[a,b) &= \{\, x \in \mathbb{R} \mid a \le x < b \,\} \\
(a,b] &= \{\, x \in \mathbb{R} \mid a < x \le b \,\}
\end{aligned}$$

The three names

- **Closed**: $[a,b]$, **both** endpoints included.
- **Open**: (a,b), **neither** endpoint included.
- **Half-open**: $[a,b)$ or $(a,b]$, one of each (also called **half-closed**).

One Decision, Two Endpoints, Four Intervals

Each endpoint decides on its own, so two endpoints give $2 \times 2 = 4$ intervals. On 2 and 7:



Only the endpoints differ, every number strictly between 2 and 7 belongs to all four. And the notations agree endpoint by endpoint:

Bracket	**Dot**	**Inequality**	**The endpoint itself**
square, [ ]	filled ●	$\leq, \geq$ (bar)	**is** in the set
round, ( )	open ○	$<, >$ (no bar)	is **not** in the set

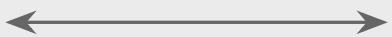
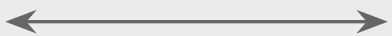
Every Interval, Three Ways

Interval	Number line	Set-builder
(a, b)		$\{x \in \mathbb{R} \mid a < x < b\}$
$(a, b]$		$\{x \in \mathbb{R} \mid a < x \leq b\}$
$[a, b)$		$\{x \in \mathbb{R} \mid a \leq x < b\}$
$[a, b]$		$\{x \in \mathbb{R} \mid a \leq x \leq b\}$
(a, ∞)		$\{x \in \mathbb{R} \mid x > a\}$
$[a, \infty)$		$\{x \in \mathbb{R} \mid x \geq a\}$
$(-\infty, b)$		$\{x \in \mathbb{R} \mid x < b\}$
$(-\infty, b]$		$\{x \in \mathbb{R} \mid x \leq b\}$
$(-\infty, \infty)$		$\mathbb{R}$

∞ is **not a number**, just shorthand for "without end," so its side always takes a **round** bracket.

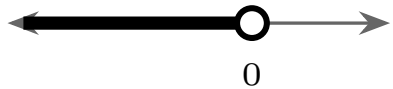
Exercise: Translating Between the Three

Try it: each row is filled in *one* way. Fill in the other three.

	Number line	**Interval**	**Inequality**	**Set-builder**
1.		$[-3, 5)$		
2.			$-1 < x \leq 6$	
3.				$\{x \in \mathbb{R} \mid x \geq 4\}$
4.				

Watch each endpoint separately, they are independent decisions.

Exercise: Solution

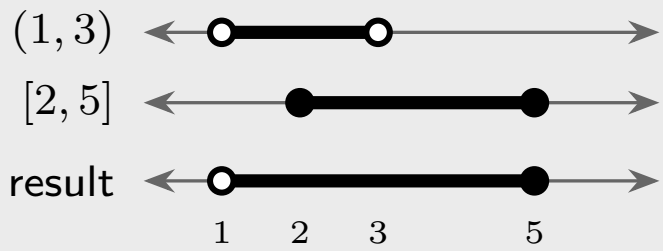
	Number line	Interval	Inequality	Set-builder
1.		$[-3, 5)$	$-3 \leq x < 5$	$\{x \in \mathbb{R} \mid -3 \leq x < 5\}$
2.		$(-1, 6]$	$-1 < x \leq 6$	$\{x \in \mathbb{R} \mid -1 < x \leq 6\}$
3.		$[4, \infty)$	$x \geq 4$	$\{x \in \mathbb{R} \mid x \geq 4\}$
4.		$(-\infty, 0)$	$x < 0$	$\{x \in \mathbb{R} \mid x < 0\}$

#3 and #4 are unbounded, so the infinite side takes a **round** bracket, there is no endpoint there to include. In #4 the open dot at 0 is what makes it $<$ rather than $\leq$.

Union of Intervals

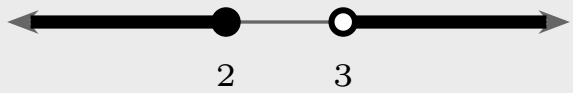
Draw both, then read off everything covered by **either** one.

Bounded: $(1, 3) \cup [2, 5] = (1, 5]$



They overlap, so they merge into one stretch: open at 1, closed at 5.

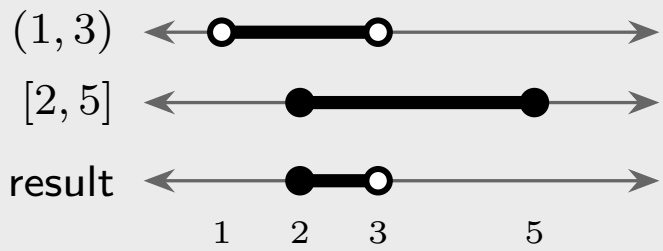
Unbounded: $(-\infty, 2] \cup (3, \infty)$, a union that does *not* merge



No overlap, so nothing merges, the union is genuinely two pieces, and this is already its simplest form. An "abnormal lab value" set looks exactly like this: too low, or too high.

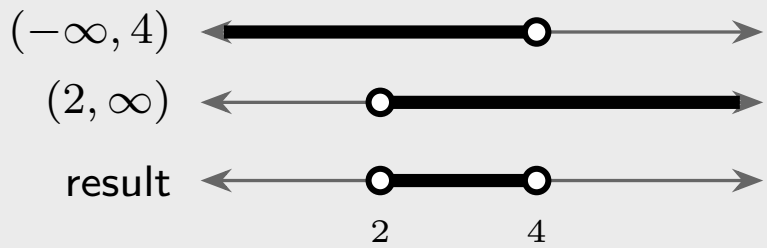
Intersection of Intervals

Bounded: $(1, 3) \cap [2, 5] = [2, 3)$



Only the overlap survives. 2 lies in **both**, so it is included; 3 is in neither, so it is not.

Unbounded: $(-\infty, 4) \cap (2, \infty) = (2, 4)$

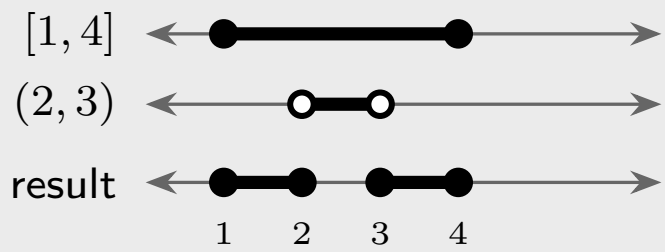


Two infinite sets can meet in a bounded one. Neither endpoint is in both, so both stay open.

Difference and Complement

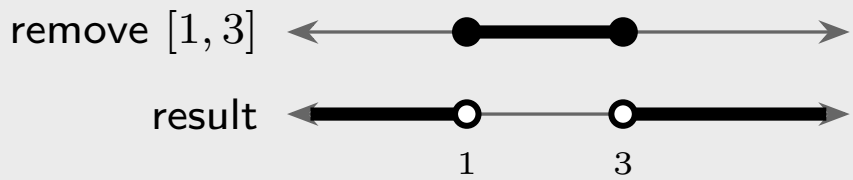
$A \setminus B$ keeps what is in A but **not** in B; the **complement** is the case $\mathbb{R} \setminus A$.

Bounded: $[1,4] \setminus (2,3) = [1,2] \cup [3,4]$



Removing an *open* middle leaves two *closed* pieces, 2 and 3 were never in $(2,3)$, so they were never removed.

Unbounded: $\mathbb{R} \setminus [1,3] = (-\infty, 1) \cup (3, \infty)$



Removing a *closed* interval leaves *open* ends: 1 and 3 were inside, so they are gone.

Watch the endpoints, every mistake happens there.

Ordered Pairs and Tuples

When Order Matters

In a set, order is irrelevant: $\{2, 5\} = \{5, 2\}$. Sometimes that is exactly wrong.

Ordered pair

An **ordered pair** (a, b) is a collection of two elements in which **the order matters**:

$$(a, b) = (c, d) \quad \Longleftrightarrow \quad a = c \text{ and } b = d$$

The one you already know: points on an xy graph

Every point you have ever plotted is an ordered pair, first entry the x-coordinate, second the y-coordinate. So $(2, 5)$ means *right 2, up 5*, while $(5, 2)$ means *right 5, up 2*: two different places from the same two numbers.

Notation warning

$(2, 5)$ is also how we write an open interval. **Context tells you which is meant**, a pair of coordinates, or a stretch of the number line.

Tuples

n-tuple

An n-**tuple** $(a_1, a_2, \ldots, a_n)$ generalizes the ordered pair to any finite number of entries, order still mattering.

- A **2-tuple** is an ordered pair: (a, b).
- A **3-tuple** (triple): (a, b, c).
- A **4-tuple**: (a, b, c, d). And so on.

Two n-tuples are equal exactly when they agree entry by entry: $a_i = b_i$ for every i.

Examples

- $(2, 5) \neq (5, 2)$, order matters.
- $(1, 2, 3) \neq (3, 2, 1)$, same reason.
- $(x, y, z) = (4, -2, 7)$ locates a point in three-dimensional space.

Why Tuples Matter Here

They are the bridge to the rest of the course

- **Coordinates:** every point on a graph is a tuple.
- **Relations and functions (Unit 1-3):** a relation is literally a *set of ordered pairs* $(input, output)$.
- **Vectors (Unit 2):** a vector in $\mathbb{R}^n$ is an n-tuple of numbers.
- **Data:** one row of a data table, one patient's measurements, is a tuple. A whole dataset is a set of tuples.

The takeaway

A **set** answers "which things?" A **tuple** answers "which things, in which slots?" Nearly all of data science is built out of the two.

Review

Review

What we covered

1. A **set** is a collection of elements; $\{x \mid \ldots\}$ describes one by a rule.
2. $A \subseteq B$ means every element of A is in B; $\emptyset$ is a subset of everything.
3. $A \cup B$, $A \cap B$, $A \setminus B$, $\bar{A}$ are the set versions of OR, AND, AND NOT, NOT, **De Morgan** included.
4. **Venn diagrams** organize the regions; $|A \cup B| = |A| + |B| - |A \cap B|$ counts them.
5. $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; rational decimals terminate or repeat, irrational ones do not.
6. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are all **countable** ($\aleph_0$); Cantor's diagonal shows $\mathbb{R}$ is strictly bigger.
7. **Intervals** are stretches of the real line; brackets record the endpoints.
8. **Tuples** are collections where order matters, points, data rows, vectors.

Notebooks: `U1-2_Sets-1_Operations`, `-2_VennDiagrams`, `-3_NumberSystems`, `-4_Intervals`